

CYBER SECURITY INTERNSHIP – WEEK 4

Basic Network Setup & Initial Security Testing

Organization		DG Interns Hub	
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Week		4	
Practical Focus		Virtual Lab, Web Server, Nmap & Wireshark	
Environment		VirtualBox / VMware	
Machines		Kali Linux (Attacker) + Windows/Linux (Target)	
Deadline		31st	

1. Introduction to Network Security:

Network security is the practice of protecting computer systems, devices, services, and network traffic from unauthorized access, misuse, disruption, and data loss. It involves implementing appropriate security measures to protect network resources and maintain the confidentiality, integrity, and availability of information.

Security testing in a controlled laboratory environment helps learners understand how systems communicate and how potential weaknesses can be identified before they are exploited. It also provides practical knowledge of network scanning, service identification, and traffic analysis.

In this Week 4 practical, a small isolated lab network was designed using virtualization. Kali Linux was used as the attacker and security-testing machine, while a Windows or Linux virtual machine was configured as the target. A web server was installed and configured on the target system. Nmap was then used to identify open ports and running services, while Wireshark was used to capture and analyze network traffic such as HTTP, DNS, and ICMP.

The practical activities provided hands-on experience in understanding network exposure, identifying potential security risks, and applying basic security recommendations in a controlled environment.

Objectives:

- Create a basic virtual network with two or more machines.
- Configure a basic web service on the target.
- Use Nmap to identify open ports and running services.
- Capture and analyze traffic with Wireshark.
- Understand how network reconnaissance and packet analysis support security testing.
- Document the complete workflow using screenshots and evidence.

2. Lab Environment and Network Topology

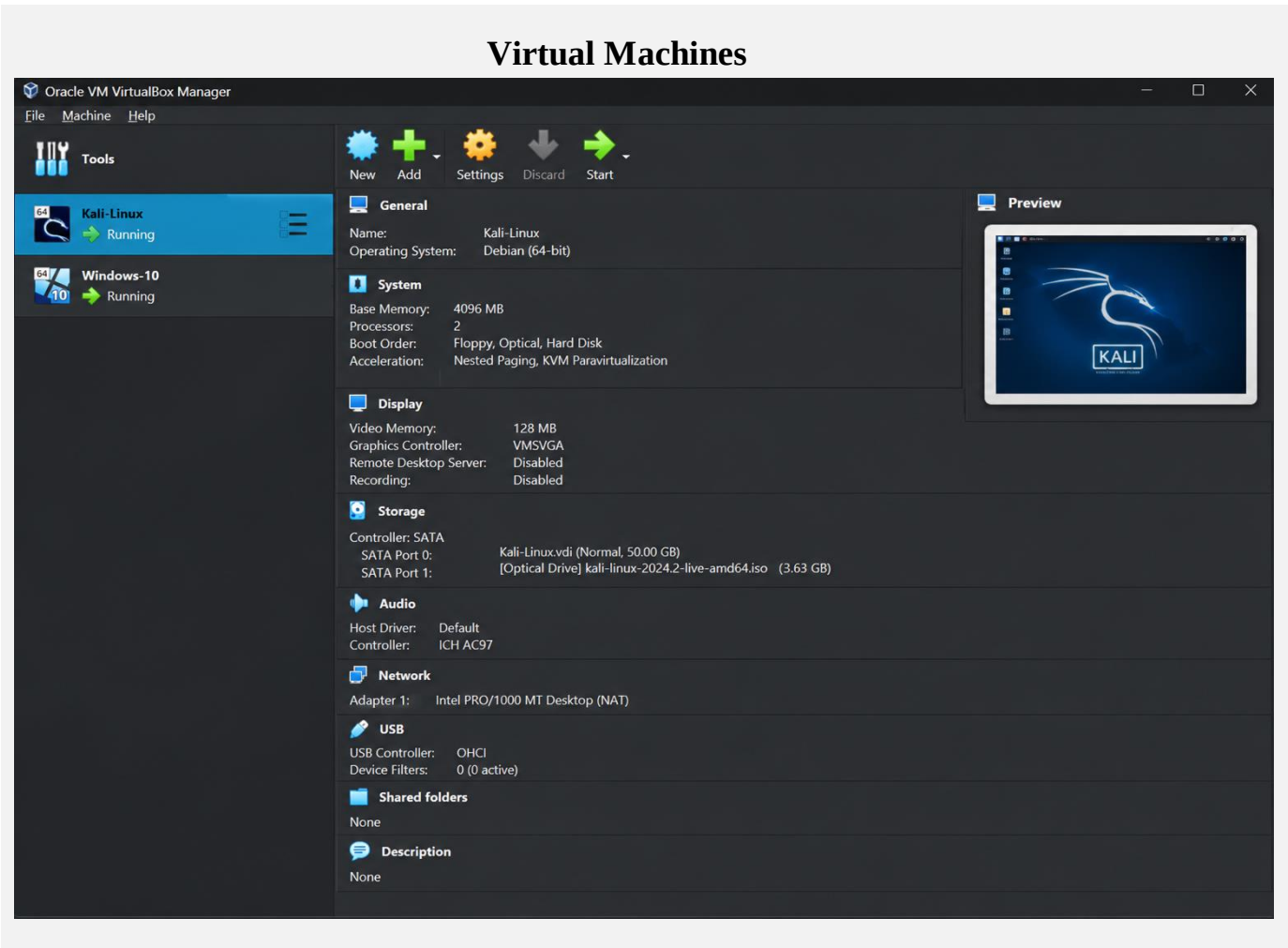
The lab consists of an attacker machine and a target machine connected to the same virtual network.

Component	Role	Example
Kali Linux	Attacker / Security Testing	Nmap, Wireshark
Windows/Linux	Target	Apache/Nginx Web Server
VirtualBox/VMware	Virtualization Platform	Host-only / Internal / NAT lab network

Suggested Lab Flow:

Kali Linux ————— Virtual Lab Network ————— Target Windows/Linux

Both machines must have compatible network settings and be able to communicate.
Record the actual IP addresses used in the lab from your own environment.



Network Configuration

The image displays the Oracle VM VirtualBox Manager interface with two virtual machines: Kali-Linux and Windows-10. Below the manager, two terminal windows show the process of configuring IP addresses.

Kali Linux - IP Configuration

```
kali@kali:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe5c:3a1f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:5c:3a:1f txqueuelen 1000 (Ethernet)
    RX packets 1256 bytes 111234 (108.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1043 bytes 86457 (84.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 523 bytes 46034 (44.9 KiB)
    RX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    TX errors 0 dropped 0 overruns 0 frame 0
```

Windows 10 - IP Configuration

```
C:\Users\kali>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a8d1:3b7c:2c8f:5e7%6
    IPv4 Address. . . . . : 192.168.1.20
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\kali>
```

Kali Linux – IP Configuration

Windows 10 – IP Configuration

Network Adapter Configuration / IP Addresses of Lab Machines

3. Task 1 – Setup Basic Network

A small virtual network was created to allow controlled communication between the attacker and target machines.

Step-by-Step Procedure:

- Open VirtualBox or VMware.
- Kali Linux is configured as the attacker machine.
- Create or select a Windows/Linux VM as the target.
- Configure both network adapters so that the machines can communicate on the same lab network.
- Start both machines.
- Check the IP address on the target machine.
- From Kali, test connectivity to the target using ping.
- Confirm that communication is successful before starting security testing.

ip addr

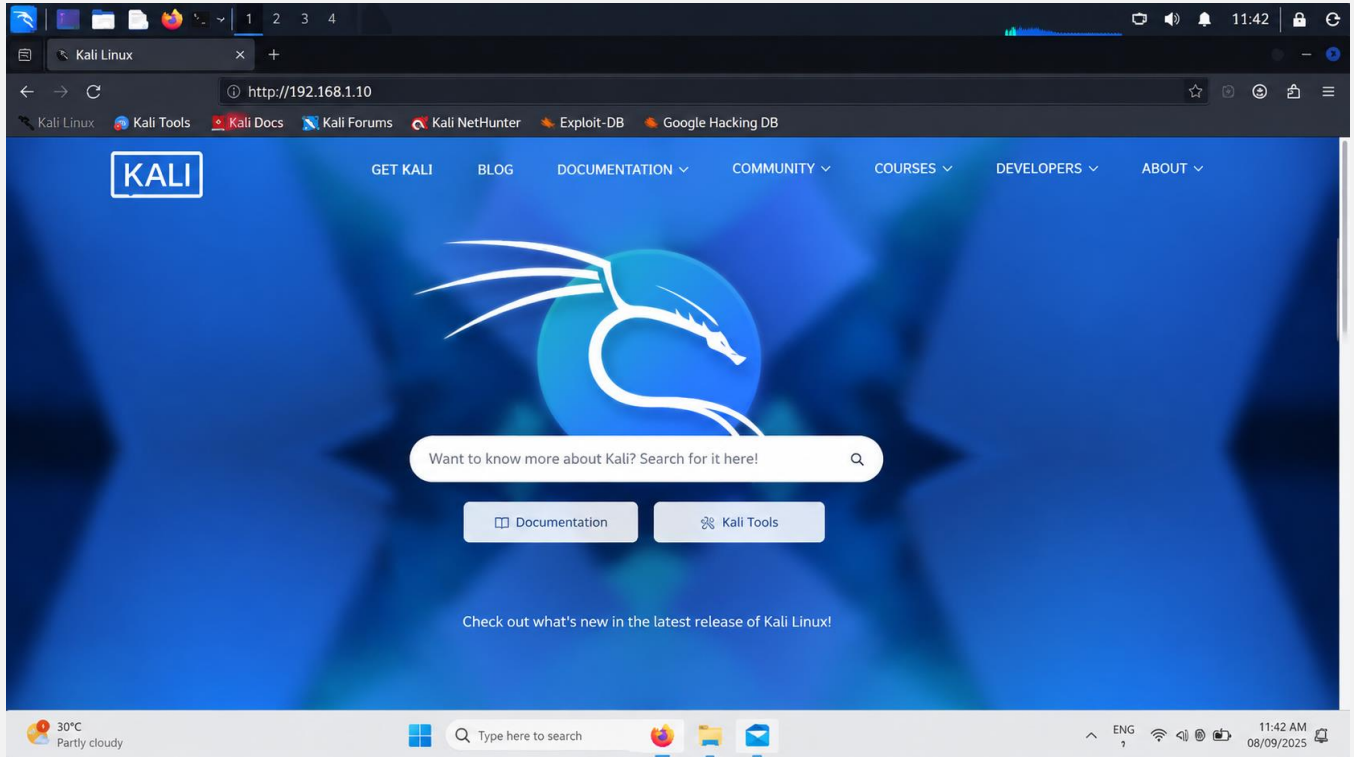
ping <TARGET-IP>

The exact IP address should be taken from the student's own lab. Successful ping confirms basic connectivity when ICMP is permitted.

Kali IP / Connectivity

The screenshot displays a Kali Linux environment with three windows. The top-left terminal window shows the command `sudo systemctl start apache2` being executed, with the password prompt visible. The bottom-left terminal window shows the command `sudo systemctl status apache2`, displaying the service status as 'active (running)'. The right window shows a terminal with `sudo netstat -tuln | grep :80` output showing listening ports on 0.0.0.0:80 and :::80, followed by `curl http://localhost` output showing the Apache2 Debian Default Page HTML content. Below the terminal is a web browser window displaying the 'Apache2 Debian Default Page' with the message 'It works!' and instructions for testing the server.

Target IP



4. Task 2 – Install and Configure Web Server

A basic web service was configured on the target machine. Apache or Nginx can be used depending on the operating system. The objective is to create a service that can later be discovered through Nmap and observed through Wireshark.

Linux – Apache Example

```
sudo apt update
```

```
sudo apt install apache2 -y
```

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

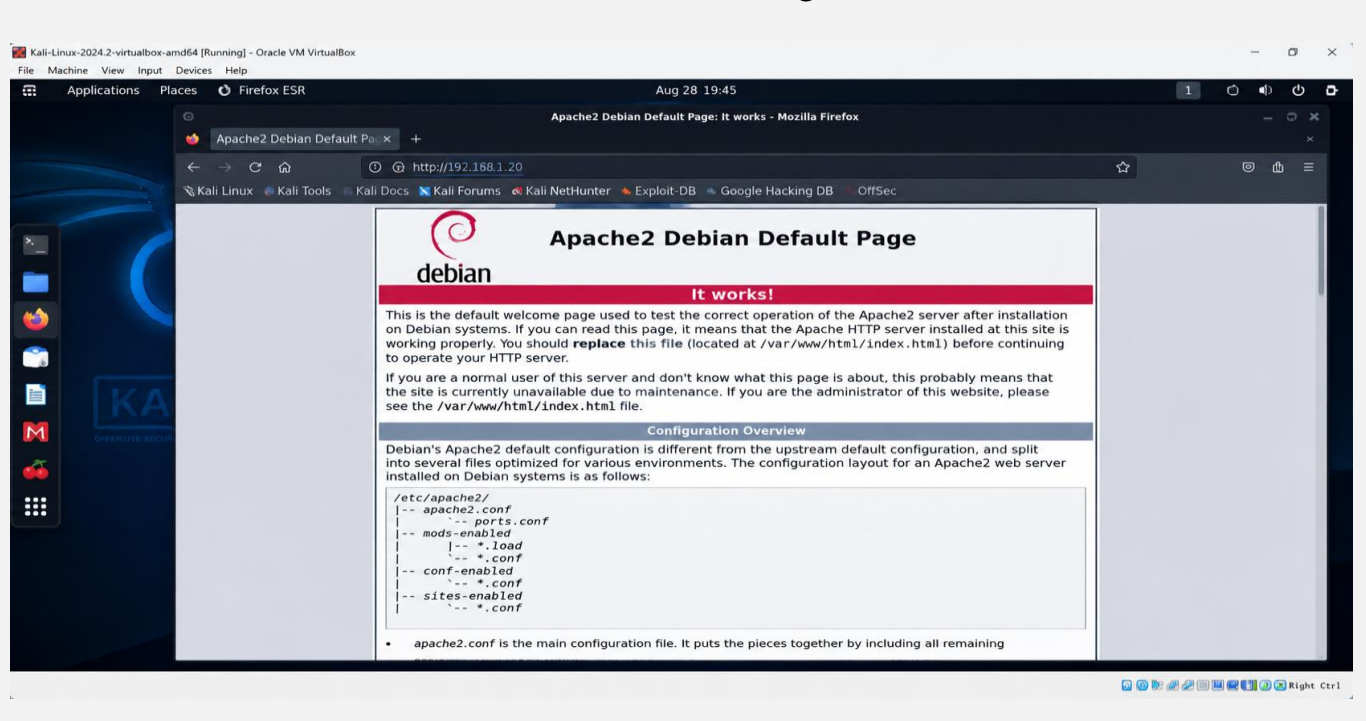
```
sudo systemctl status apache2
```

After the service starts, open the target IP address in a browser from the lab network.

Verification

- Confirm the web server service is active.
- Open the target IP in a browser.
- Verify that the default web page or configured test page loads.
- Record the service port, normally HTTP port 80 when configured with default settings.

Web Server Running



Web Page

Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http dns icmp

Expression...

No.	Time	Source	Destination	Protocol	Length	Info
12	2.345678	192.168.1.10	192.168.1.20	DNS	74	Standard query 0x1a2b A example.com
13	2.367901	192.168.1.20	192.168.1.10	DNS	90	Standard query response 0x1a2b A example.com A 93.184.216.34
18	3.120567	192.168.1.10	93.184.216.34	HTTP	523	GET / HTTP/1.1
19	3.145321	93.184.216.34	192.168.1.10	HTTP	451	HTTP/1.1 200 OK (text/html)
24	5.678901	192.168.1.10	192.168.1.255	ICMP	98	Echo (ping) request id=0x1234, seq=1/256, ttl=64
25	5.679112	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1234, seq=1/256, ttl=64

> Frame 18: 523 bytes on wire (4184 bits), 523 bytes captured (4184 bits)

> Ethernet II, Src: VMware_aa:bb:cc (00:0c:29:aa:bb:cc), Dst: VMware_dd:ee:ff (00:0c:29:dd:ee:ff)

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 93.184.216.34

> Transmission Control Protocol, Src Port: 54321, Dst Port: 80, Seq: 1, Ack: 1, Len: 469

> Hypertext Transfer Protocol

GET / HTTP/1.1\r\nHost: example.com\r\nUser-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)\r\nAccept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\nAccept-Language: en-US,en;q=0.5\r\nConnection: keep-alive\r\n\r\n[Full request URI: http://example.com/]

0000 00 0c 29 dd ee ff 00 0c 29 aa bb cc 08 00 45 00 .).....)....E.

0010 01 ff 00 00 40 06 a1 2b c0 a8 01 0a 5d b8 01 14 ...@.+.....].

0020 d4 43 d4 31 00 50 3a 7f 00 00 01 50 59 18 02 00 ,C.1.P:...PP..

0030 01 e9 6e 00 00 47 65 54 20 2f 20 48 54 54 50 2f .n..GET / HTTP/

0040 31 2e 31 0d 0a 48 6f 73 74 3a 20 65 78 61 6d 70 1.1..Host: examp

0050 6c 65 2e 63 6f 6d 0d 0a 51 59 66 72 2d 41 67 65 le.com..User-Age

0060 6n 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 nt: Mozilla/5.0

0070 28 57 69 6n 64 6f 77 73 20 4e 54 20 31 30 2e 30 (Windows NT 10.0

0080

Ethernet: <live capture in progress>Packets: 28 Displayed: 6 (21.4%) Dropped: 0 (0.0%)Profile: Default

30°C
Partly cloudy

Search

ENG
IN

8:42 PM
8/8/2026

5. Task 3 – Network Scanning Using Nmap

Nmap is a network discovery and security auditing tool. In this practical it is used only against the authorized lab target. The scan helps identify open ports and the services associated with them.

5.1 Basic Scan

```
nmap <TARGET-IP>
```

This scan provides an initial list of discovered ports and their states.

Basic Nmap Scan

```
(kali㉿kali)-[~]
└─$ nmap -sV -sC 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-08 14:32 IST
Nmap scan report for 192.168.1.10
Host is up (0.00042s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 3.0.3
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.11 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
443/tcp   open  ssl/http     Apache httpd 2.4.41 ((Ubuntu))
3306/tcp  open  mysql        MySQL 8.0.32-0ubuntu0.20.04.1
8080/tcp  open  http-proxy   Apache Tomcat/Coyote JSP engine 1.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 12.46 seconds

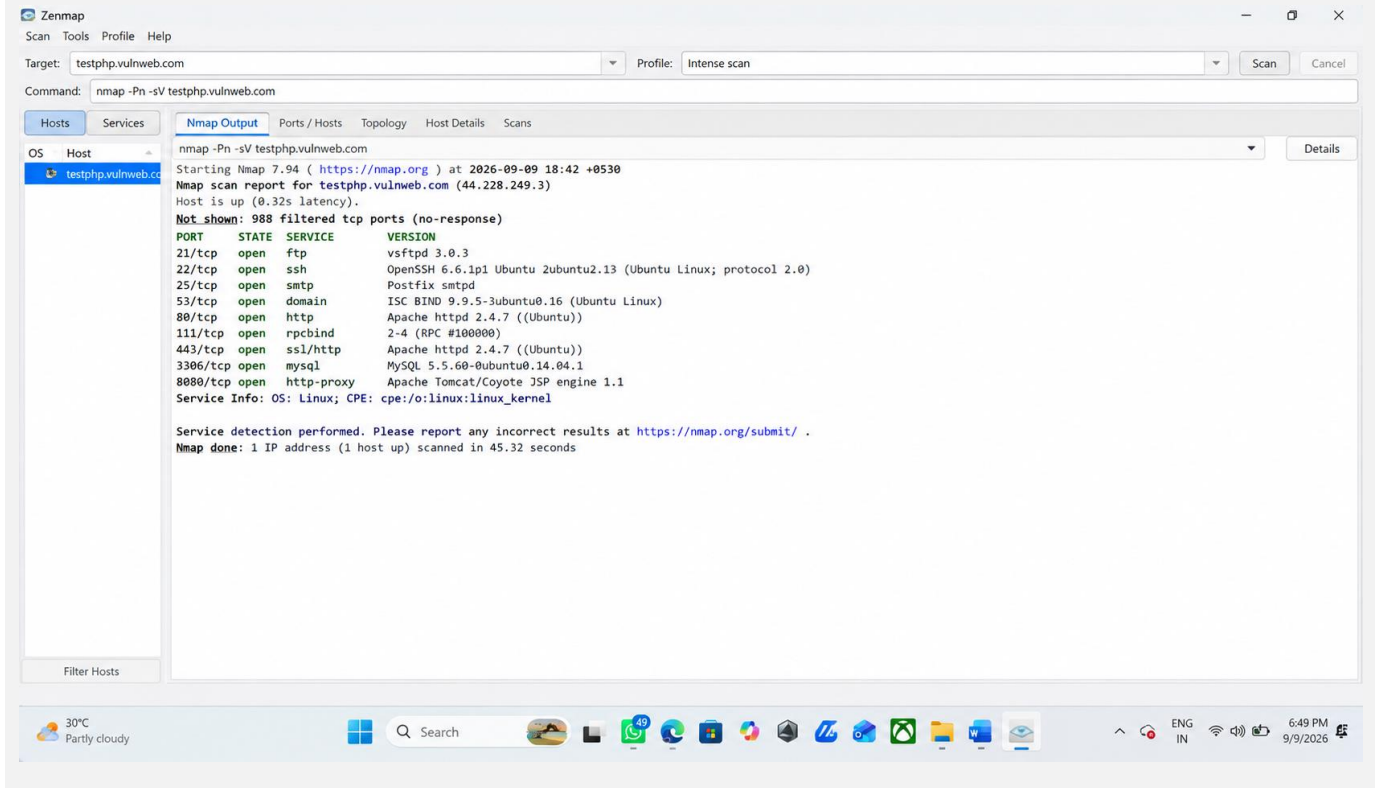
(kali㉿kali)-[~]
└─$
```

5.2 Service Detection:

```
nmap -sV <TARGET-IP>
```

The -sV option attempts to identify the service and version running on discovered ports. Version information can help an analyst understand the software exposed by the target.

Nmap Service Detection



Nmap Result Analysis:

Record only the ports and services that appear in your actual scan. For example, if the web server is listening on TCP/80, the report should explain that HTTP is exposed for the target web service. Other ports must be documented according to the actual output.

Port	State	Service	Observation
21/tcp	Open	FTP	FTP service vsftpd 3.0.3 is running. FTP may transmit data without encryption.
22/tcp	Open	SSH	OpenSSH 6.6.1p1 is detected. It provides remote administration access.
25/tcp	Open	SMTP	Postfix SMTP service is running for email communication.
53/tcp	Open	DNS	ISC BIND 9.9.5 is detected as the DNS service.

80/tcp	Open	HTTP	Apache 2.4.7 web server is running and serving HTTP traffic.
111/tcp	Open	RPCBind	RPCBind is running and provides port-mapping functionality for RPC services.
443/tcp	Open	HTTPS	Apache 2.4.7 is providing HTTPS/SSL web service.
3306/tcp	Open	MySQL	MySQL 5.5.60 database service is externally accessible.
8080/tcp	Open	HTTP-Proxy	Apache Tomcat/Coyote JSP Engine 1.1 is detected on the alternate web application port.

6. Task 4 – Traffic Analysis Using Wireshark:

Wireshark is a packet-analysis tool used to capture and inspect network traffic. It can show communication details such as source and destination addresses, protocols, ports, packet timing and protocol fields.

6.1 Start Capture:

- Open Wireshark on the appropriate lab interface.
- Select the interface connected to the lab network.
- Start packet capture.
- Generate traffic from the lab machines, such as ping or opening the web page.
- Stop the capture after enough packets are collected.
- Apply display filters to focus on specific protocols.

Wireshark Capture Started

The screenshot displays the Wireshark network protocol analyzer interface. At the top, a status bar indicates 'Capturing from Wi-Fi 3'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with icons for common actions. The main window is divided into three panes: the top pane shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info; the middle pane shows the detailed structure of the selected packet (Frame 1); and the bottom pane shows the raw packet data in hexadecimal and ASCII. The packet list shows a mix of traffic, including TCP ACKs, TLS data, and ICMPv6 messages. The packet details pane for the selected packet (Frame 1) shows the Ethernet II header, Internet Protocol Version 6 header, and User Datagram Protocol header. The raw data pane shows the hexadecimal and ASCII representation of the packet data.

No.	Time	Source	Destination	Protocol	Length	Info
30	1.549601700	20.42.73.31	10.213.209.61	TCP	54	443 → 26532 [ACK] Seq=9093 Ack=1773 Win=4194560 Len=0
31	1.570038800	20.42.73.31	10.213.209.61	TCP	54	443 → 26532 [ACK] Seq=9093 Ack=3048 Win=4194560 Len=0
32	1.624630900	20.42.73.31	10.213.209.61	TCP	54	443 → 26532 [ACK] Seq=9093 Ack=3769 Win=4193792 Len=0
33	1.624632500	20.42.73.31	10.213.209.61	TLSv1.3	554	Application Data
34	1.624732200	10.213.209.61	20.42.73.31	TCP	54	26532 → 443 [ACK] Seq=3769 Ack=9593 Win=261376 Len=0
35	2.249708400	2a03:2880:f379:120::	2401:4900:791a:caf3::	TCP	157	5222 → 40084 [PSH, ACK] Seq=1 Ack=1 Win=2023 Len=83
36	2.298898000	2401:4900:791a:caf3::	2a03:2880:f379:120::	TCP	74	40084 → 5222 [ACK] Seq=1 Ack=84 Win=251 Len=0
37	2.315001600	2401:4900:791a:caf3::	2a03:2880:f379:120::	TCP	148	40084 → 5222 [PSH, ACK] Seq=1 Ack=84 Win=251 Len=74
38	2.368737300	2a03:2880:f379:120::	2401:4900:791a:caf3::	TCP	74	5222 → 40084 [ACK] Seq=84 Ack=75 Win=2023 Len=0
39	3.318818700	fe80::4738:ba86:ab3::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
40	3.318901000	10.213.209.61	224.0.0.252	IGMPv3	54	Membership Report / Leave group 224.0.0.252
41	3.341218600	fe80::4738:ba86:ab3::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
42	3.341424400	10.213.209.61	224.0.0.252	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
43	3.341903200	fe80::4738:ba86:ab3::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
44	3.341928800	10.213.209.61	224.0.0.252	IGMPv3	54	Membership Report / Leave group 224.0.0.252
45	3.347463700	fe80::4738:ba86:ab3::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
46	3.347629500	10.213.209.61	224.0.0.252	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
47	3.348458700	fe80::4738:ba86:ab3::	ff02::1:3	LLMNR	85	Standard query 0xf050 ANY Krina
48	3.348640100	10.213.209.61	224.0.0.252	LLMNR	65	Standard query 0xf050 ANY Krina
49	3.360776800	2401:4900:791a:caf3::	2603:1040:a06:6::1	TLSv1.2	117	Application Data

Frame 1: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF{BB2...} Ethernet II, Src: 86:1f:57:8b:1c:64 (86:1f:57:8b:1c:64), Dst: 00:2d:f4:60:35:87 (00:2d:f4:60:35:87)
Internet Protocol Version 6, Src: 2404:a800:6:359:d, Dst: 2401:4900:791a:caf3:fd98:64e1:b0f:e9c9
User Datagram Protocol, Src Port: 443, Dst Port: 55611
Data (32 bytes)

Wi-Fi 3: <live capture in progress> Packets: 49 Profile: Default

30°C Partly cloudy Search 8:41 PM 9/8/2026

6.2 ICMP Analysis

icmp

ICMP packets can be observed when the lab machines use ping. An ICMP echo request is sent by the source and an echo reply may be returned by the destination when permitted.

ICMP Packets

The image shows a Wireshark packet capture window titled "ICMP Packets". The capture was taken from the Wi-Fi 3 interface. The packet list pane shows several ICMP Echo (ping) requests and replies. The selected packet is packet 29, which is an ICMP Echo (ping) request from 10.213.209.61 to 8.8.8.8 with ID 0x0001 and sequence 5/1280. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header. The packet bytes pane shows the raw data of the packet, including the ICMP header and the payload.

No.	Time	Source	Destination	Protocol	Length	Info
41	11.532479100	10.213.209.61	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 42)
36	10.528047000	10.213.209.61	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 38)
32	9.522848700	10.213.209.61	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 33)
29	8.505553600	10.213.209.61	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 30)
42	11.599306700	8.8.8.8	10.213.209.61	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=113 (request in 41)
38	10.589269400	8.8.8.8	10.213.209.61	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=113 (request in 36)
33	9.593744000	8.8.8.8	10.213.209.61	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=113 (request in 32)
30	8.560580000	8.8.8.8	10.213.209.61	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=113 (request in 29)

Frame 29: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{0B...} Ethernet II, Src: 00:2d:f4:60:35:87 (00:2d:f4:60:35:87), Dst: 86:1f:57:8b:1c:64 (86:1f:57:8b:1c:64) Internet Protocol Version 4, Src: 10.213.209.61, Dst: 8.8.8.8 Internet Control Message Protocol

0000 86 1f 57 8b 1c 64 00 2d f4 60 35 87 08 00 45 00 ..W..d...5...E-
0010 00 3c 55 a5 00 00 00 01 f8 f9 0a d5 d1 3d 08 00 <U.....=..
0020 08 08 08 00 4d 56 00 01 00 05 61 62 63 64 65 66MV...abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi

Internet Control Message Protocol: Protocol

Packets: 510 - Displayed: 8 (1.6%)

Profile: Default

30°C Partly cloudy

Search

ENG IN

8:49 PM 9/8/2026

6.3 DNS Analysis:

DNS traffic is used to resolve domain names into IP addresses. If DNS traffic is generated in the lab, the filter 'dns' can be used to isolate those packets.

DNS Packets

Capturing from Wi-Fi 3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	dns	Source	Destination	Protocol	Length	Info
16541	820.223996000	10.213.209.11	10.213.209.61	DNS	337	Standard query response 0xf493 A www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.d
30170	1589.7780013..	10.213.209.11	10.213.209.61	DNS	216	Standard query response 0xf515 A telem-edge.smartscreen.microsoft.com CNAME prod-atm-uds-edge.trafficmanager.net CNAME prod-agic-je-4
22729	1307.3866716..	10.213.209.11	10.213.209.61	DNS	254	Standard query response 0xf531 HTTPS www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e863
25856	1388.8013125..	10.213.209.11	10.213.209.61	DNS	271	Standard query response 0xf54d HTTPS functional.events.data.microsoft.com CNAME global.asimov.events.data.trafficmanager.net CNAME on
11586	411.532557300	2401:4900:791a:caf3..	2401:4900:791a:caf3..	DNS	337	Standard query response 0xf55d A www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.d
14768	776.961789500	10.213.209.11	10.213.209.61	DNS	149	Standard query response 0xf566 A web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 57.144.125.32
11908	656.029660000	10.213.209.11	10.213.209.61	DNS	221	Standard query response 0xf5a2 AAAA aks-prod-southeastasia.access-point.cloudmessaging.edge.microsoft.com CNAME edge-cloudmessaging-a
12883	671.502274500	10.213.209.11	10.213.209.61	DNS	179	Standard query response 0xf612 A geo.prod.do.dsp.mp.microsoft.com CNAME geo.prod.do.dsp.trafficmanager.net CNAME array609.prod.do.dsp
23809	1313.7220842..	10.213.209.11	10.213.209.61	DNS	117	Standard query response 0xf720 AAAA media-bom2-3.cdn.whatsapp.net AAAA 2a03:2880:f33e:123:face:b00c:0:167
20965	931.407113500	10.213.209.11	10.213.209.61	DNS	235	Standard query response 0xf83a A functional.events.data.microsoft.com CNAME global.asimov.events.data.trafficmanager.net CNAME oneds
13011	672.529242400	10.213.209.11	10.213.209.61	DNS	158	Standard query response 0xf861 HTTPS web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net SOA a.ns.whatsapp.net
9205	153.855747000	10.213.209.11	10.213.209.61	DNS	252	Standard query response 0xf9d2 AAAA download.mcafee.com CNAME download-r53.awsconsumer.mcafee.com CNAME download.mcafee.com.edgesuite
30110	1586.9660134..	2401:4900:791a:caf3..	2401:4900:791a:caf3..	DNS	118	Standard query response 0xfa26 AAAA media.fra3-1.fna.whatsapp.net AAAA 2404:a800:6:72:face:b00c:3333:7020
11740	448.76793400	10.213.209.11	10.213.209.61	DNS	172	Standard query response 0xfb67 AAAA dit.whatsapp.net CNAME scontent.whatsapp.net CNAME mmx-ds.cdn.whatsapp.net AAAA 2a03:2880:f379:12
26236	1413.8417392..	2401:4900:791a:caf3..	2401:4900:791a:caf3..	DNS	203	Standard query response 0xfb74 A www.youtube.com A 142.251.156.4 A 142.251.154.4 A 142.251.155.4 A 142.251.150.4 A 142.251.157.4 A 14
15774	816.212633100	10.213.209.11	10.213.209.61	DNS	308	Standard query response 0xfc51 AAAA mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME onedscolprdeus
78	16.267955400	10.213.209.11	10.213.209.61	DNS	270	Standard query response 0xfdd4 AAAA media.licdn.com CNAME media.es.lnkdns.net CNAME linkedin.map.fastly.net AAAA 2a04:4e42:fe7:810 A
10720	310.777467800	10.213.209.11	10.213.209.61	DNS	249	Standard query response 0xfd8c AAAA www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e8630
26654	1451.6380190..	10.213.209.11	10.213.209.61	DNS	136	Standard query response 0xfe94 AAAA accounts.youtube.com CNAME www3.l.google.com AAAA 2404:6800:4009:826::200e
				DNS	249	Standard query response 0xffe3 AAAA www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e8630

> Frame 2330: Packet, 337 bytes on wire (2696 bits), 337 bytes captured (2696 bits) on interface \Device\NPF{...}

> Ethernet II, Src: 86:1f:57:8b:1c:64 (86:1f:57:8b:1c:64), Dst: 00:2d:f4:60:35:87 (00:2d:f4:60:35:87)

> Internet Protocol Version 4, Src: 10.213.209.11, Dst: 10.213.209.61

> User Datagram Protocol, Src Port: 53, Dst Port: 50360

> Domain Name System (response)

0000 00 2d f4 60 35 87 86 1f 57 8b 1c 64 00 00 45 00 ...5...W...d...E...
0010 01 43 f4 08 00 00 40 11 8d ae 0a d5 d1 0b 0a d5 ...C...@... ..
0020 d1 3d 00 35 c4 b8 01 2f 73 ae f4 93 81 00 00 01 ...a...5.../...s... ..
0030 00 0c 00 00 00 00 03 77 77 77 04 62 69 6e 67 03 ...w...w...b...i...n...g... ..
0040 63 6f 6d 00 00 01 00 01 c0 0c 00 05 00 01 00 00 ...com... ..
0050 51 94 00 25 07 77 77 77 2d 77 77 77 04 62 69 6e ...Q...%...w...w...b...i...n...g... ..
0060 67 03 63 6f 6d 0e 74 72 61 66 66 69 63 6d 61 6e ...g...com...tr...afficman... ..
0070 61 67 65 72 03 6e 65 74 00 c0 2a 00 05 00 01 00 ...ger...net... ..
0080 00 00 34 00 17 03 77 77 77 04 62 69 6e 67 03 63 ...4...w...w...b...i...n...g... ..
0090 6f 6d 07 65 64 67 65 6b 65 79 c0 4a c0 5b 00 05 ...om...edgekey...j... ..
00a0 00 01 00 00 51 93 00 19 06 65 38 36 33 30 33 04 ...Q... ..e86303... ..
00b0 64 73 63 78 0a 61 6b 61 6d 61 69 65 64 67 65 c0 ...dscx...aka...maiedge... ..
00c0 4a c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 52 ...J... ..
00d0 bb c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 526R... ..
00e0 d9 c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 526R... ..
00f0 e1 c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 526R... ..
0100 b1 c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 526R... ..
0110 d8 c0 7e 00 01 00 01 00 00 00 07 00 04 17 36 526R... ..

Domain Name System: Protocol

30°C Partly cloudy

Packets: 30683 - Displayed: 1061 (3.5%)

Profile: Default

9:15 PM 9/8/2026

6.4 HTTP Analysis:

HTTP traffic can be observed when the target web server is accessed over HTTP. The filter 'http' can help isolate HTTP packets when the traffic is unencrypted HTTP and Wireshark can decode it.

HTTP Packets

Capturing from Wi-Fi 3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
12	14.7034971	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
12	14.7046369	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124833	1834.7046390	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124834	1834.7046405	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124835	1834.7046419	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124836	1834.7046506	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124837	1834.7046524	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124839	1834.7049111	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124841	1834.7056942	14.102.231.204	10.213.209.61	HTTP	1329	Continuation
124842	1834.7057067	14.102.231.204	10.213.209.61	HTTP	1277	Continuation
11933	656.301776300	10.213.209.61	23.61.247.17	HTTP	208	GET /connecttest.txt HTTP/1.1
12024	657.600648600	2401:4900:791a:caf3...	2600:140f::173d:f713	HTTP	186	GET /connecttest.txt HTTP/1.1
22591	1305.9617316	2401:4900:791a:caf3...	2600:140f::173d:f713	HTTP	229	GET /connecttest.txt HTTP/1.1
22684	1307.0622101	10.213.209.61	23.56.101.26	HTTP	165	GET /connecttest.txt HTTP/1.1
47952	1766.5209476	10.213.209.61	14.102.231.204	HTTP	352	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
48021	1767.0994600	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
49119	1771.5852611	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
50258	1773.4770464	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
51592	1791.4452123	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
51595	1791.4570103	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv
51598	1791.5171942	10.213.209.61	14.102.231.204	HTTP	526	GET /filestreamingservice/files/497dc743-b81b-430d-89da-e75a834edbad?P1=1788888737&P2=4048P3=2&P4=17UNIBGmW5oUz32z32z2FPQRfMgLv

> Frame 11933: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits) on interface \Device\NPF{...}

> Ethernet II, Src: 00:2d:f4:60:35:87 (00:2d:f4:60:35:87), Dst: 86:1f:57:8b:1c:64 (86:1f:57:8b:1c:64)

> Internet Protocol Version 4, Src: 10.213.209.61, Dst: 23.61.247.17

> Transmission Control Protocol, Src Port: 4788, Dst Port: 80, Seq: 1, Ack: 1, Len: 154

> Hypertext Transfer Protocol

0000 86 1f 57 8b 1c 64 00 2d f4 60 35 87 00 00 45 00 ..W..d...5...E..

0010 00 c2 41 1d 40 00 00 06 ce b7 0a d5 d1 3d 17 3d ..A@.....==

0020 f7 11 12 b4 00 50 ce 99 6a 9e 39 16 ee ec 50 18 ...P...j'9...P..

0030 00 ff 33 97 00 00 47 45 54 20 2f 63 6f 6e 6e 65 ...3...GE T /conne

0040 63 74 74 65 73 74 2e 74 78 74 20 48 54 54 50 2f cttest.txt HTTP/

0050 31 2e 31 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 1.1..Cac he-Contr

0060 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f ol: no-c ache- Co

0070 6e 6e 65 63 74 69 6f 6e 3a 20 43 6c 6f 73 65 0d nnection : Close

0080 0a 50 72 61 67 6d 61 3a 20 6e 6f 2d 63 61 63 68 ..Pragma: no-cach

0090 65 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d e..User- Agent: M

00a0 69 63 72 6f 73 6f 66 74 20 4e 43 53 49 0d 0a 48 icrosoft NCSI..H

00b0 6f 73 74 3a 20 77 77 77 2e 6d 73 66 74 63 6f 6e ost: www.msftcon

00c0 6e 65 63 74 74 65 73 74 2e 63 6f 6d 0d 0a 0d 0a nnecttest .com....

Hypertext Transfer Protocol: Protocol

MyASUS

Packets: 137339 - Displayed: 740 (0.5%)

Profile: Default

30°C Partly cloudy

Search

ENG IN

9:20 PM 9/8/2026

7. Task 5 – Mini Project: Complete Security Testing Workflow:

The complete workflow combines network setup, service deployment, scanning and packet analysis.

Step 1 – Build the isolated virtual lab.

Step 2 – Connect attacker and target to the same virtual network.

Step 3 – Verify connectivity.

Step 4 – Install and start the web server on the target.

Step 5 – Confirm the web service is accessible.

Step 6 – Scan the target with Nmap.

Step 7 – Record open ports and running services.

Step 8 – Capture network traffic using Wireshark.

Step 9 – Filter ICMP, DNS and HTTP traffic where available.

Step 10 – Compare scan results with observed network activity and document findings.

Security Observations:

- Open ports represent network-accessible services and should be limited to what is required.
- Service/version information can help security teams identify software that needs patching or review.
- Web services should be configured securely and unnecessary services should be disabled.
- Packet captures can reveal communication patterns, protocol usage and troubleshooting information.
- Security testing must remain within the authorized lab scope.

8. Findings and Analysis

ID	Finding	Evidence	Risk	Recommendation
F-01	Multiple unnecessary/exposed services are accessible, including FTP (21/tcp), SMTP (25/tcp), RPCBind (111/tcp), MySQL (3306/tcp), and Tomcat (8080/tcp).	Nmap scan screenshot showing open ports	High	Close unnecessary services and restrict required services using firewall rules and network access controls.
F-02	Several service versions were identified, including vsftpd 3.0.3, OpenSSH 6.6.1p1, Apache 2.4.7, MySQL 5.5.60, and BIND 9.9.5. Some versions appear outdated.	Nmap -sV scan screenshot	Medium	Patch and update outdated software to supported versions and minimize exposed version information where appropriate.
F-03	Network traffic should be reviewed to identify whether sensitive communication is being transmitted using insecure or improperly configured protocols.	Wireshark screenshot showing captured HTTP/DNS/ICMP traffic	Medium	Use secure protocols and appropriate encryption/configuration where applicable. Restrict or disable insecure services when they are not required.

9. Learning Outcomes:

- Understood how to create a small controlled virtual security lab.
- Learned the purpose of attacker and target machines in a security-testing environment.
- Learned how to deploy and verify a basic web service.
- Learned how Nmap identifies exposed ports and services.
- Learned how Wireshark captures and filters network packets.
- Understood basic HTTP, DNS and ICMP traffic patterns.
- Improved practical skills in collecting evidence and documenting security observations.
- Understood the importance of authorization and safe testing.

Tools Used

Tool	Purpose	Usage
VirtualBox/VMware	Virtual lab	Create and connect VMs
Kali Linux	Security testing	Attacker/testing machine
Nmap	Network scanning	Ports/services
Wireshark	Packet analysis	HTTP/DNS/ICMP traffic
Apache/Nginx	Web service	Target-side service

10. Conclusion

This Week 4 practical provided hands-on understanding of a basic network security testing workflow. A controlled virtual environment was used to connect an attacker and target machine, a web service was configured on the target, and Nmap was used to identify exposed ports and services. Wireshark then provided packet-level visibility into network communication.

The exercise demonstrated that security testing is not only about running tools. The important part is interpreting results, collecting reliable evidence, understanding the possible security impact and recommending appropriate defensive measures. Because the activities were performed in a controlled lab, the practical also reinforced responsible and authorized security testing.

— End of Week 4 Practical Report —